Speaking and listening with a telephone 7a

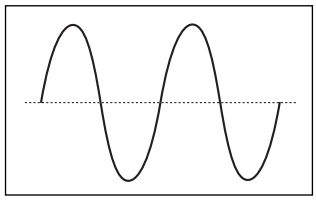
Early telephones contained a carbon microphone in a handheld mouthpiece along with a separate earpiece, dialing mechanism and other parts. The microphone changed sound waves into varying electric currents. The earpiece converted varying electric currents from another telephone back into sound waves.

High pitch sounds have high frequencies and result in electric currents that change direction (alternate) quickly (with high frequency). Deep or low pitch sounds with their lower frequencies end up as electric currents with a lower frequency. Soft sounds have vibrations with small amplitudes, while loud sounds have vibrations with large amplitudes.

A carbon microphone is a small box with two metal plates – one fixed and the other free to vibrate, each connected by wire with other parts of an electrical circuit. Carbon powder fills the space between the two plates. When the carbon powder is loosely packed, it conducts electricity poorly. When it is compressed, it is a better conductor of electricity.

A soft, deep sound hits the free-moving plate (diaphragm) of a carbon microphone resulting in weak and low frequency vibrations of the plate. These plate vibrations cause weak and low frequency electricity pulses in the wires leading away from the carbon microphone.

The diagram at right represents a medium pitched normal sound. The sound wave has been represented like a transverse wave in the diagram, although it is really a longitudinal wave. Transverse waves are easier to draw and differences can be shown clearly, so particular sound waves are often sketched like transverse waves.



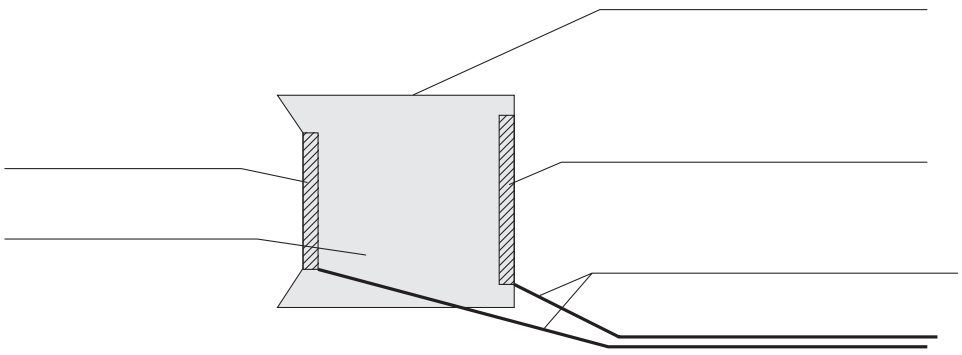
1 Draw wave diagrams in the cells of the following table. Use the diagram above as a referent for amplitude and frequency.

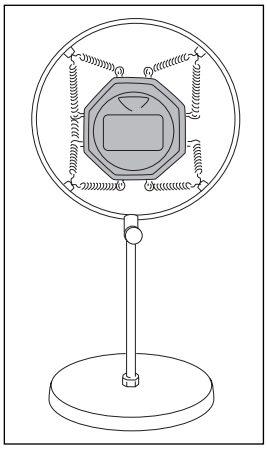
Higher pitch	Lower pitch	Quieter sound
Louder sound	Higher, louder	Deeper, quieter

2 Complete the table by describing the characteristics of particular sounds and their resulting electric current. Choose from higher or lower, higher or lower, stronger or weaker, and higher or lower.

Sound	Sound wave frequency	Sound wave amplitude	Electric pulse strength	Electric pulse frequency
Soft and deep				
Soft and high				
Loud and high				
Loud and deep				

Carbon microphone



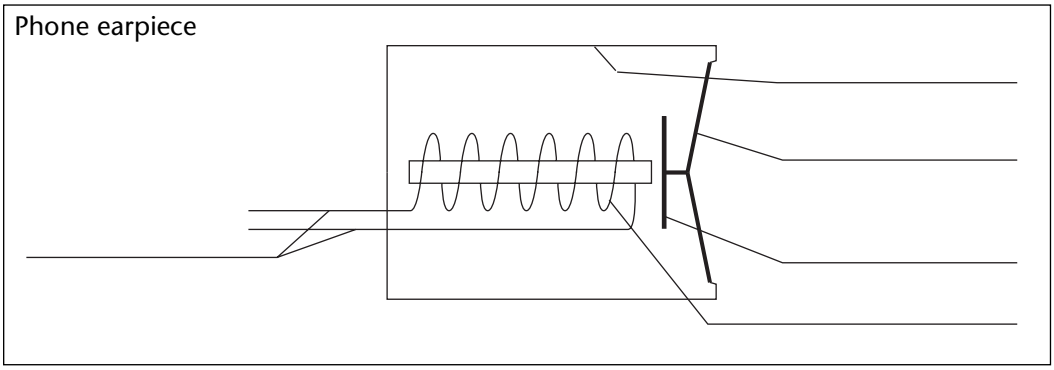


© Chris Greef, Suada Bilali and Blake Education 2006

Speaking and listening with a telephone

7b

3 Label the diagram of the carbon microphone on the previous page.



Fact 1: A strong alternating current in the wire around the iron core causes a strong varying magnetic field, particularly around the end of the iron core.

Fact 2: A strong varying electromagnetic field attracts and repels the iron plate strongly resulting in large amplitude vibrations of the plate.

Fact 3: A large amplitude movement of the diaphragm attached to the plate (usually a piece of cardboard) produces a loud sound.

Fact 4: A high frequency electric pulse results in a high frequency movement of the diaphragm and this causes a high pitch sound.

4 Label the diagram of the earpiece shown above.

5 Write a fact list for a weak, low frequency alternating current entering the earpiece.

6 Imagine you are building a telephone system made up of two phones – so there are two earpieces, two microphones, two batteries and several wires. Sketch how you would arrange these pieces. (Hint: There are actually two circuits.) Note that in mobile phones, the connecting wires have been replaced by radiowaves.

7 A high pitch sound goes from your microphone to the earpiece of a friend's phone. On a separate sheet of paper, write a paragraph or draw a flowchart explaining how this sound is transferred.

© Chris Greef, Suada Bilali and Blake Education 2006